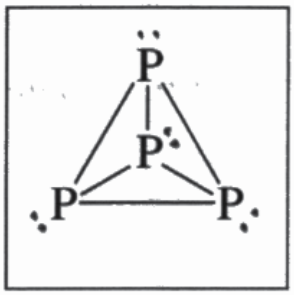
Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 3

Part A



Part B

i) ΔS° is negative because as the reaction progresses, gas molecules are used to produce solid molecules. Solids have less entropy than gases because there are less possible locations and positions of the solid molecules compared to the gas molecules. The solid molecules also move slower. So, the entropy of the system decreases and ΔS° is negative.

ii) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

The student is correct. A reaction is thermodynamically favorable when ΔG° is negative. Since ΔS° is negative, $-T\Delta S^\circ$ will be a positive value. The only way that the reaction can be thermodynamically favorable is if ΔH° is negative.

Question 3 is continued on the next page.

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Question 3

Part C

i) $q = (100.0\,g)(4.18\,J/g°C)(22.38°C - 22.00°C)$

$= 160\,J$

$= 0.16\,kJ$

ii) $\frac{0.100\,g\,P_4O_{10}}{1} \times \frac{1\,mol\,P_4O_{10}}{283.9\,g} \times \frac{1\,mol_{rxn}}{1\,mol\,P_4O_{10}} = 0.000352\,mol_{rxn}$

$\Delta H^\circ_{rxn} = \frac{-0.16\,kJ}{0.000352\,mol_{rxn}} = -450\,kJ/mol\,rxn$

Part D

ΔT would less than ΔT of the first trial. P_4O_{10} is the limiting reactant, so the moles of P_4O_{10} present determines how many reactions occur. When there is less P_4O_{10} in the calorimeter, less reactions ocurr, so less heat is transferred to the water. So, the water experiences less change in temperature.

Question 3 is continued on the next page.

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Question 3

Part E

$\frac{1}{4}P_{4(s)} + \frac{6}{4}Cl_{2(g)} \rightarrow \cancel{PCl_{3(g)}}$ $\Delta H_1^\circ = \frac{1}{4}(-1148\ kJ/mol\ rxn)$

$\cancel{PCl_{3(g)}} + Cl_{2(g)} \rightleftarrows PCl_{5(g)}$ $\Delta H_2^\circ = -88\ kJ/mol\ rxn$ +

$\frac{1}{4}P_{4(s)} + \frac{5}{2}Cl_{2(g)} \rightarrow PCl_{5(g)}$ $\Delta H_f^\circ = -375\ kJ/mol\ rxn$

Part F

i) $K_p = \frac{P_{PCl_5}}{P_{PCl_3} \cdot P_{Cl_2}} = \frac{(4.00)}{(2.00)(6.00)} = 0.333$

ii) $PCl_3 + Cl_2 \rightleftarrows PCl_5 + 88\ kJ/mol\ rxn$

K_p decreases when the temperature is raised. ΔH_2° is negative, so the reaction is exothermic. When heat is added, the reaction will shift towards the reactants and produce more PCl_3 and Cl_2. When the concentrations of PCl_3 and Cl_2 increase, K_p decreases.

Continue to Question 4.